

Chapter 5

Storyboard and UI/UX



Roles and Responsibilities in Software Development Team

- Project manager
- Product owner
- Team Lead
- Tech Lead
- **Developer**
 - Use UML to communicate complex ideas with other team members
 - Implement the project
 - Will be discussed in Chapter 6
- QA (Quality Assurance)
- TESTER
- **UI Designer**
- **UX Designer**

UX - User Experience

Don Norman: The term “UX”

<https://www.youtube.com/watch?v=9BdtGjoIN4E>

All aspects of the end-user’s **interaction** with the company, its services and its products

UX design is NOT about visuals; it focuses on the overall feel of the experience.

User experience (UX) designers design wireframe prototype of the system

UI - User Interface

Deals with the **interaction** between users and computer systems, software and applications

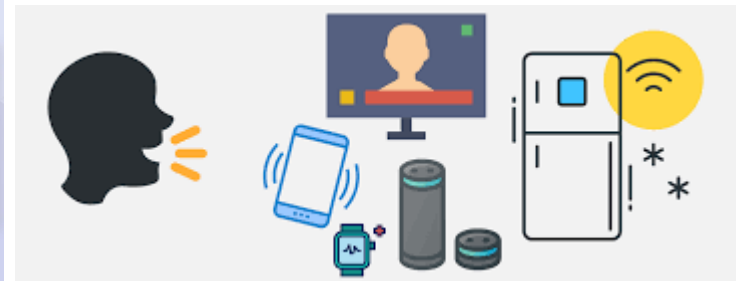
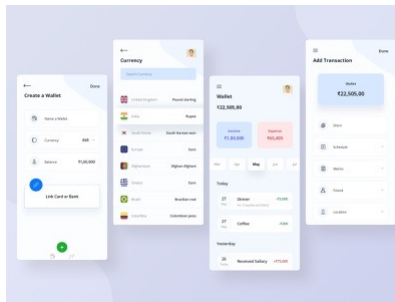
Common types of UIs

- graphical user interface
- command line interface
- voice-enabled user interface

Graphical UI includes the visual elements

- buttons
- graphics
- colour schemes
- typography
- etc.

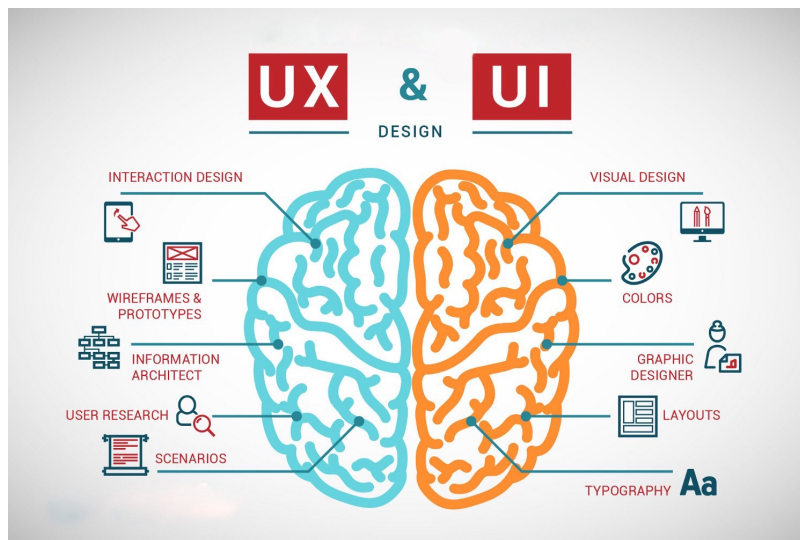
```
john@ubuntu: ~  
john@ubuntu:~$ ls  
john_directory john_file  
john@ubuntu:~$ ls -l  
total 8  
drwxrwxr-x 2 john john 40 Oct 1 11:10 john_directory  
-rw-rw-r-- 1 john john 5120 Oct 1 11:17 john_file  
john@ubuntu:~$ ls -l -h  
total 8.0K  
drwxrwxr-x 2 john john 40 Oct 1 11:10 john_directory  
-rw-rw-r-- 1 john john 5.0K Oct 1 11:17 john_file  
john@ubuntu:~$ ls -lh john_file  
-rw-rw-r-- 1 john john 5.0K Oct 1 11:17 john_file  
john@ubuntu:~$ ls -l --human-readable john_file  
-rw-rw-r-- 1 john john 5.0K Oct 1 11:17 john_file  
john@ubuntu:~$
```



UI and UX

Difference between UX and UI?

https://youtu.be/5KUNmgt_pvY



What is a Storyboard?

A storyboard is a visual representation of a film sequence.

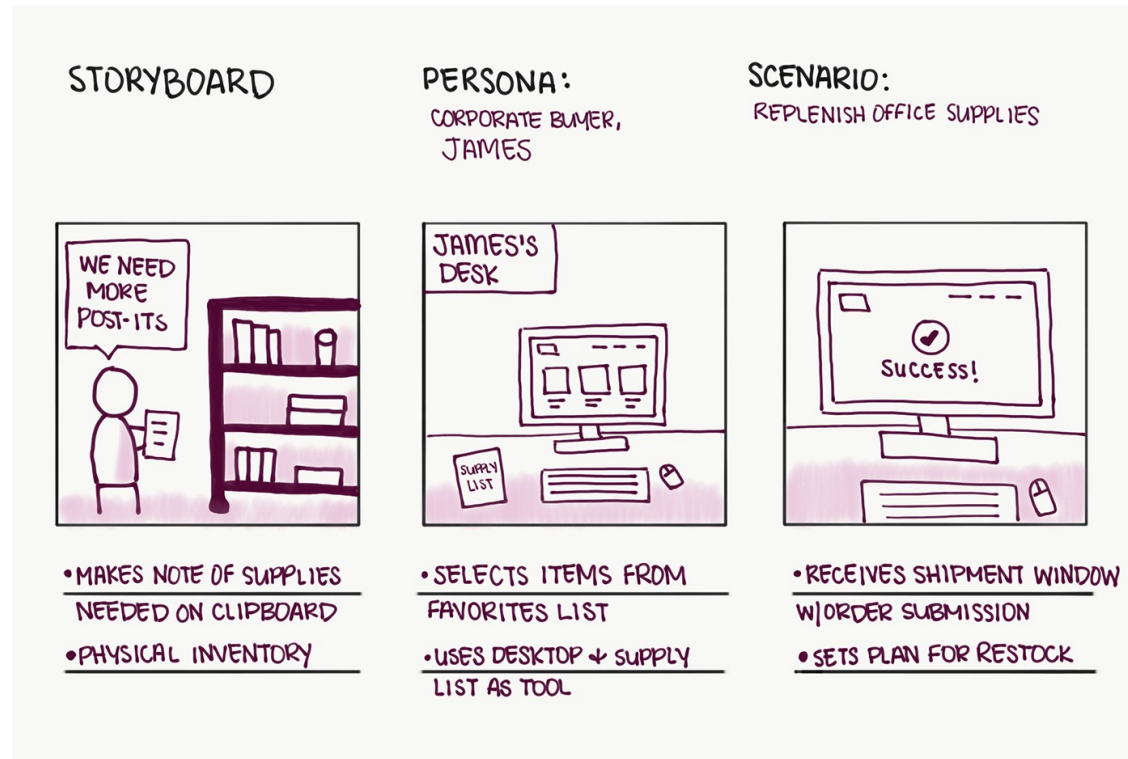
A storyboard breaks down the action into individual panels.

Storyboards were most commonly used in media or film production.

Software developers are now incorporating storyboarding into their processes.

Elements of a Storyboard

- Scenario
- Visuals
- Captions



Elements of a Storyboard

1. Scenario

Storyboards are based on a scenario or a user story.

The persona or role that corresponds to that scenario is specified at the top of the storyboard.

A short text description of the scenario is also included.

2. Visuals

Each step in the scenario is represented visually in a sequence. The image is the primary content in a storyboard. They can be sketches, illustrations, or photos.

3. Captions

Each visual has a corresponding caption. The caption describes the user's actions, environment, emotional state, device, and so on.

Captions are concise and don't typically exceed two bullet points.

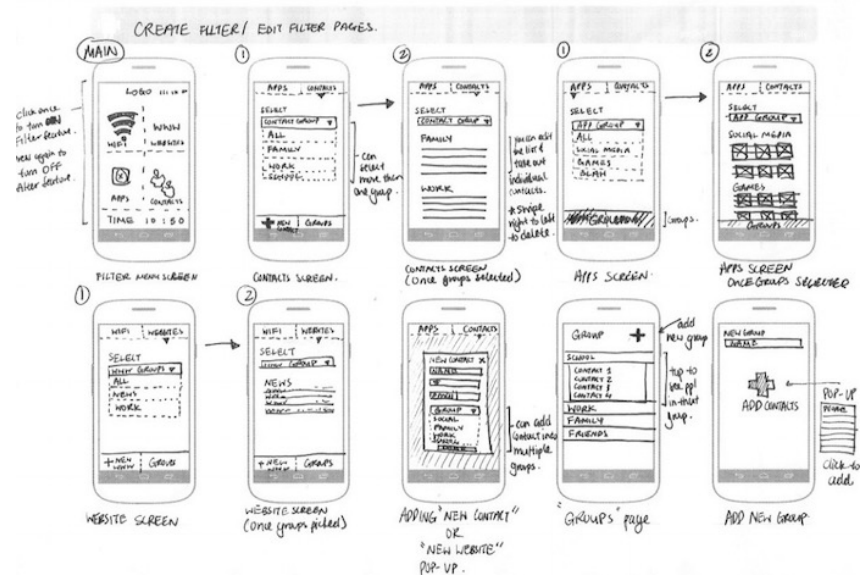
Storyboard in Software Design

Creating a storyboards forces software development team to

- think through, and
- articulate the problem faced and the possible solutions.

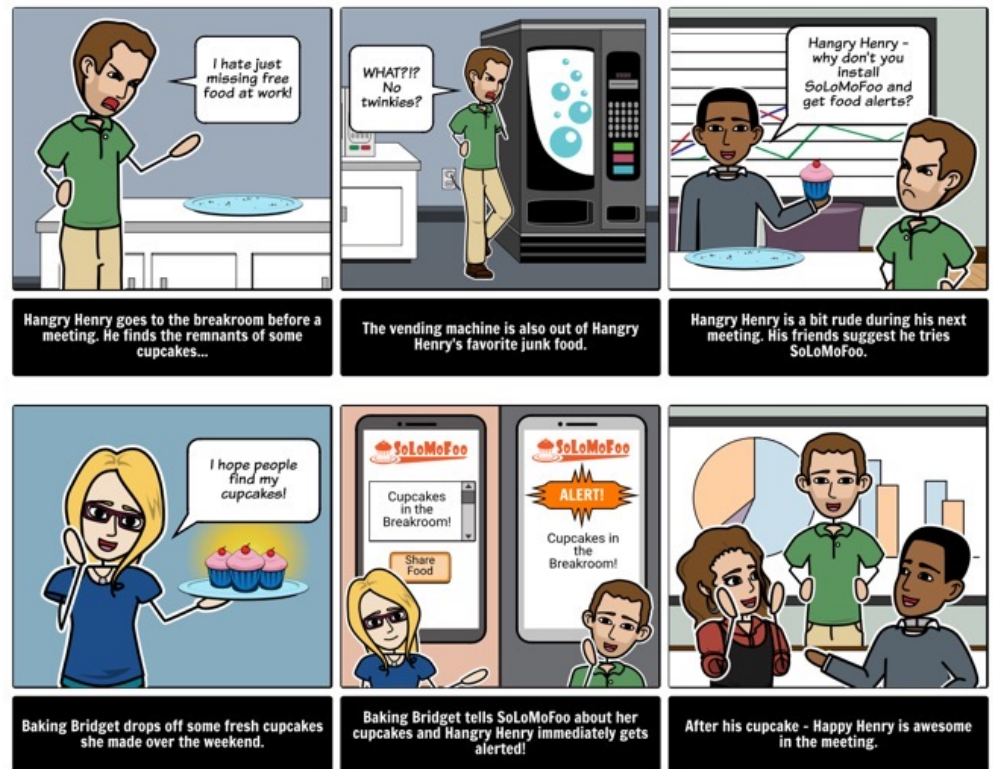
Storyboard helps to

1. Describe user's problems
2. Visualise UX ideas
3. Visualise UI design



Use Storyboards to Describe User's Problem

Storyboards help us tell stories about users.

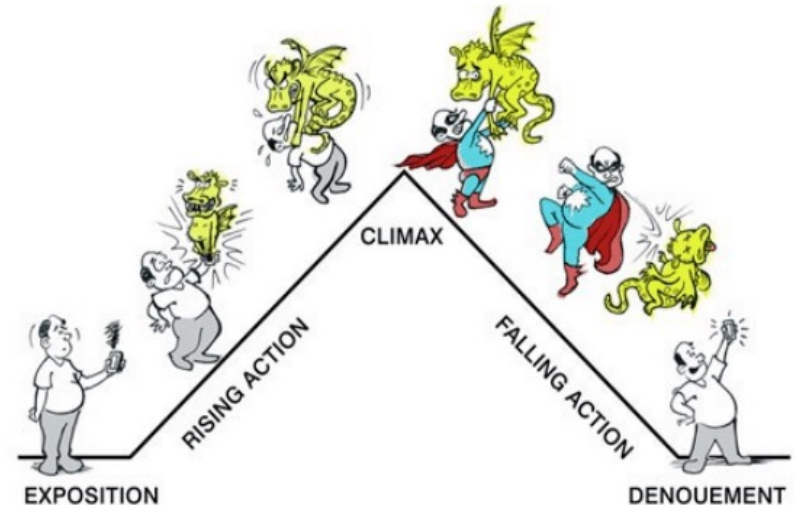


Create your own at Storyboard That

Use Storyboards to Describe User's Problem (cont'd)

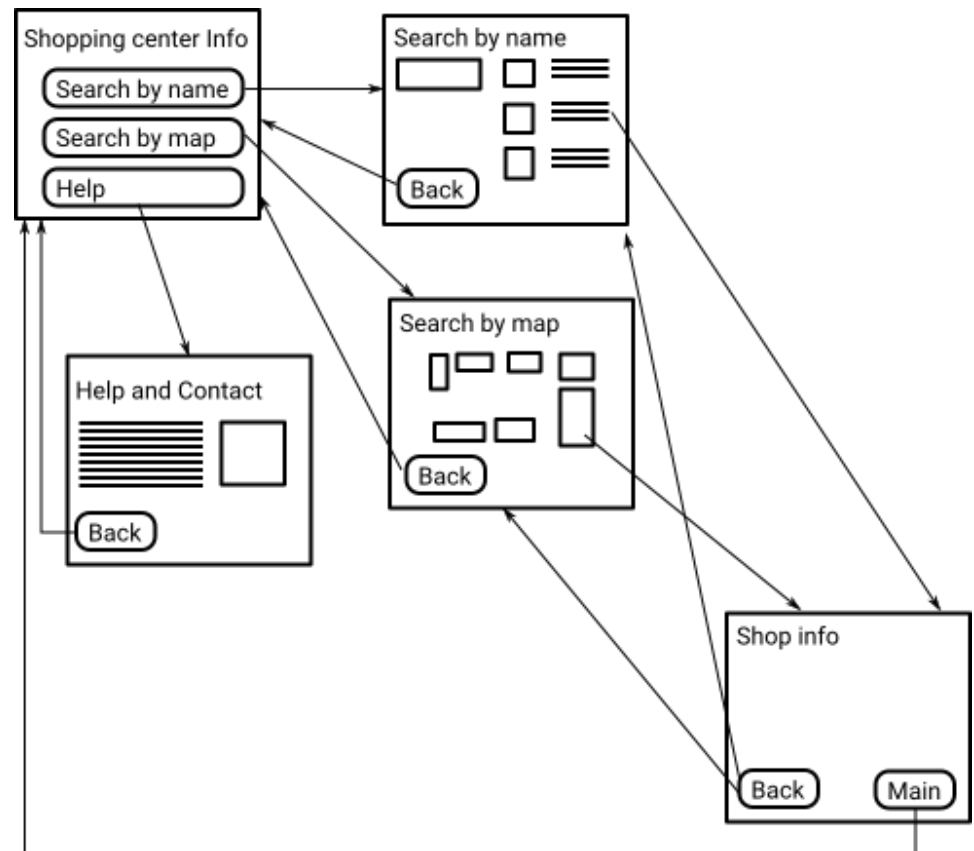
Mapping out product idea as storyboard

- What's the problem? (Exposition)
- What really sucks about it? (Rising action)
- What's the solution? (Climax)
- How does it work? (Falling action)
- What's the resolution? (Conclusion)



Storyboard in UI & UX Design

Example: Storyboard for a kiosk in a shopping centre



Types of layout structure

There are four types of page layout structure.

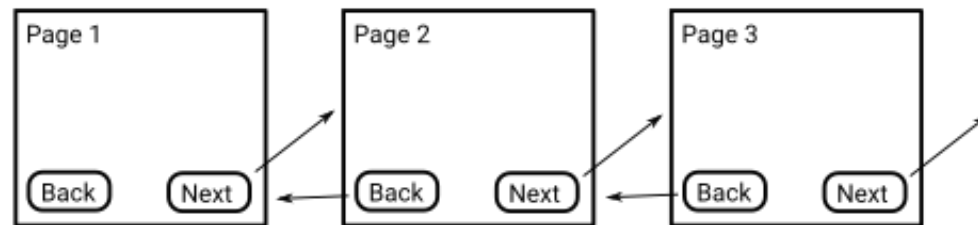
- **Linear**
- **Hierarchical**
- **Non linear**
- Composite (advanced topic which is not covered in this module)

Linear

Simple structure

There is a single main path between the screens in an orderly fashion.

This is useful when the task is sequential and fairly fixed in terms of how you will go about doing it.



Traditional e-book is a good example of this type of layout.

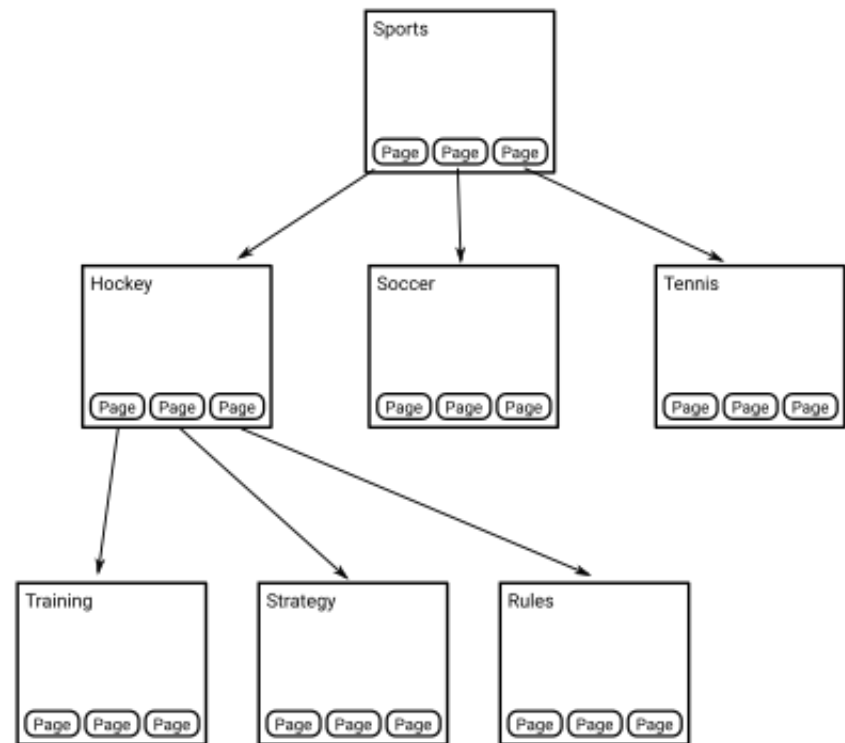
Hierarchical

There is a main page.

Beneath that are sub pages and beneath them are further sub pages and so on.

This is useful when there are decisions that need to be made and the user will go to different screens based upon those decisions.

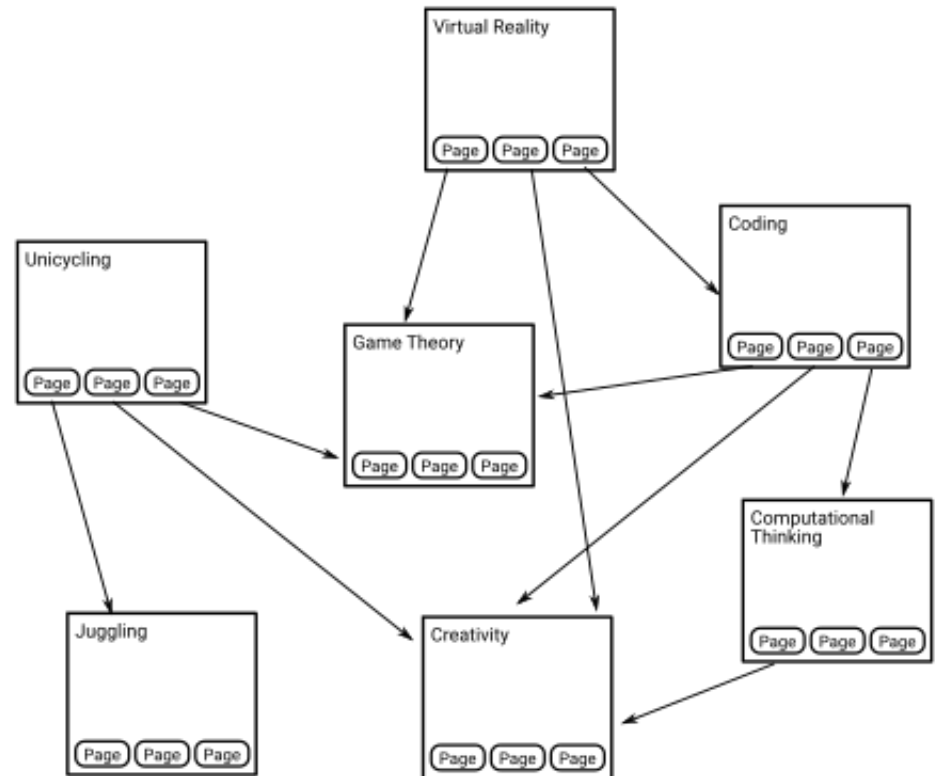
It is also useful when you have a lot of information to present which is organised categorically.



Non linear

There are links between pages in a non structured way.

The task is non linear and the user can make decisions leading to different screens.



Wikipedia is a good example of this type of layout.

UX&UI Design Case Study

Design for an app that shows food information when snapped



<https://medium.com/figma-africa/shazam-for-food-or-mobile-app-ui-design-process-2e358e853881>

Users

Lifestyle influencer /Restaurant reviewer

- They are the people we can count on to review restaurants for quality of food, ambience, service, price range and so on.

Foodie

- This person has an ardent love for food and beverages.
- A foodie seeks new food experiences as a hobby rather than simply eating out of convenience or hunger.

User problem

People eat at restaurants and events and find foods they enjoy and sometimes want information about the food they had or even information about food before making an order.

Also, tourists or people new to an environment often have trouble identifying which food is what.

With this app users can take a picture of the food and get nutritional and other information about the food displayed on the app.

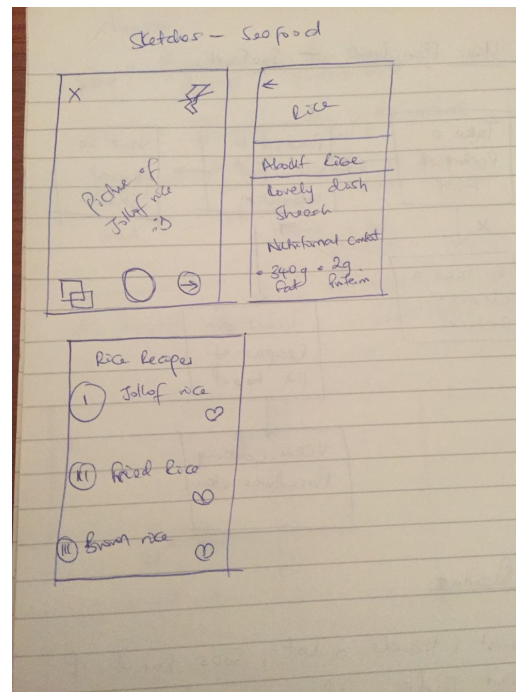
Gather user requirements

Mandatory requirements

The app should

- have a camera for user to take a picture of the food.
- be able to display content for those who are mindful of what they eat.
- allow user to find recipes for those who love to experiment.

Sketch design of the app according to user requirements



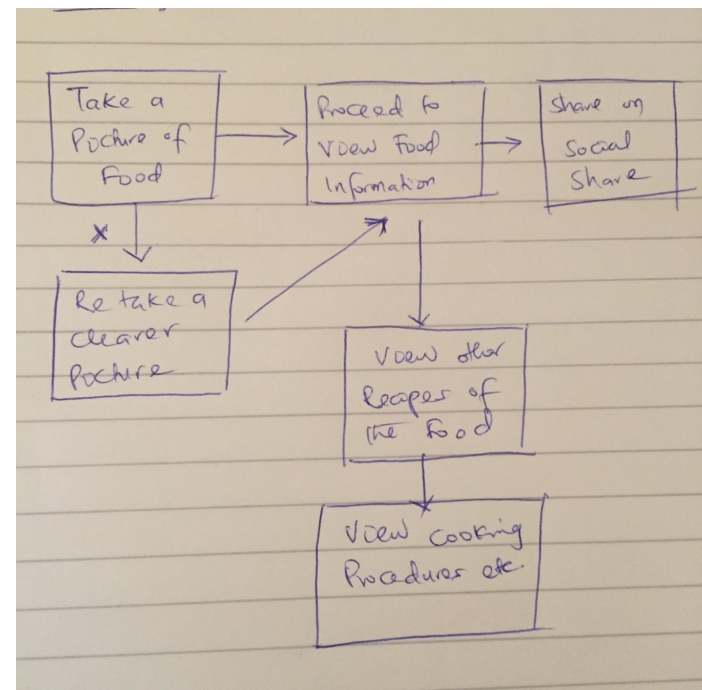
App navigation wireframe

The user arrives at an app camera

The user can take a picture of food

The app provides some additional information

- history about the food
- nutritional content
- how else the food can be eaten



More user requirements

Desirable requirement

- User accounts
- View saved food

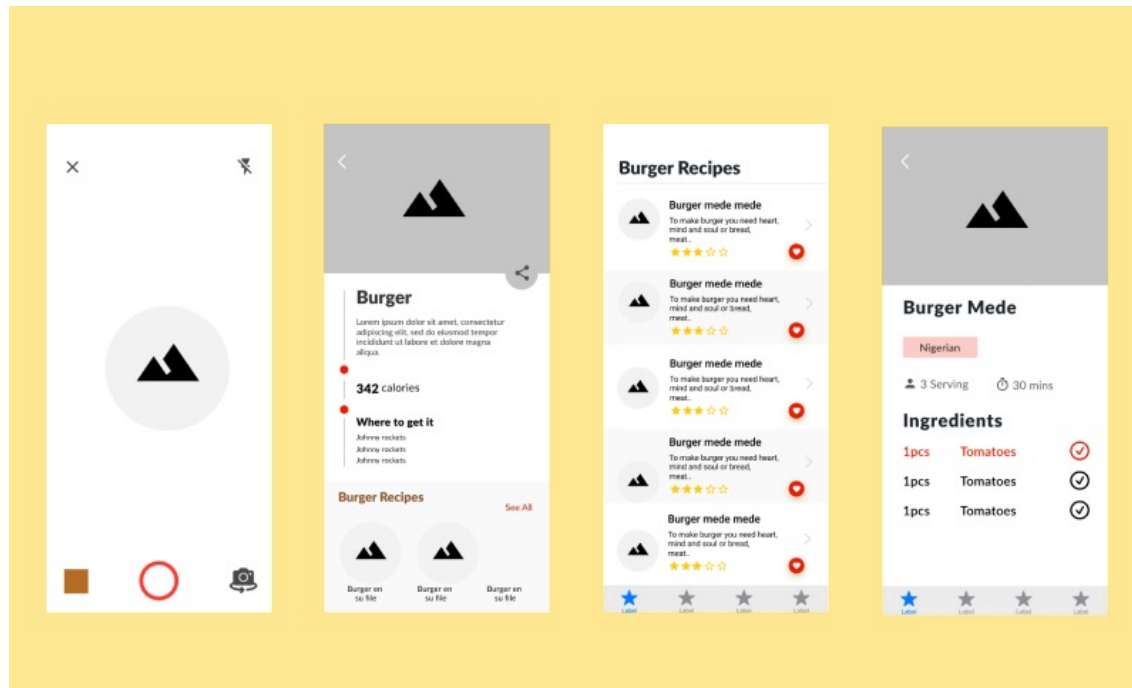
Optional requirement

- Search for restaurants to find selected food

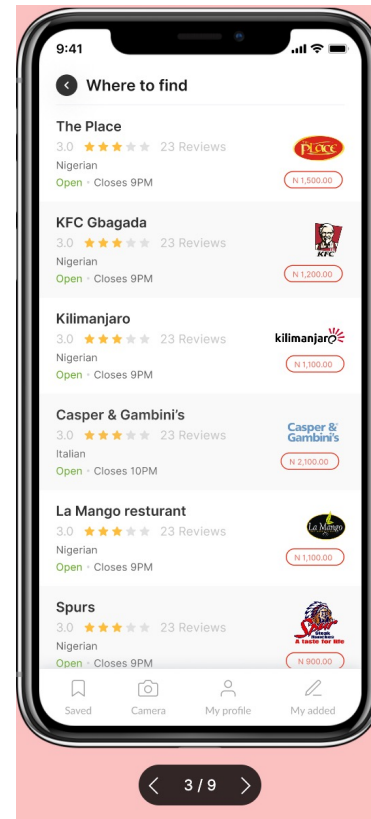
Possible future enhancement

- Navigation to the selected restaurant

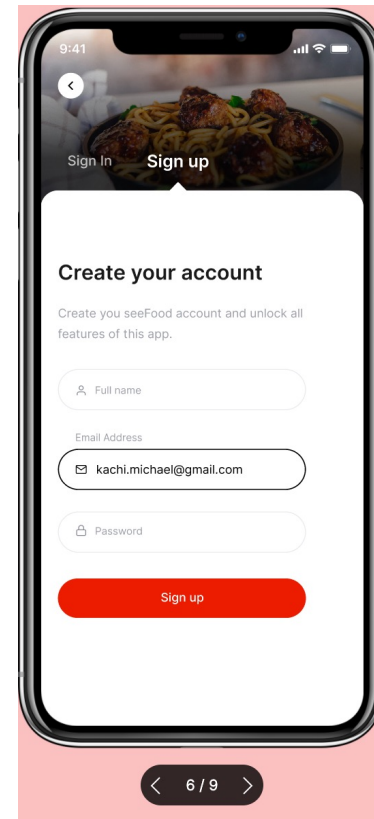
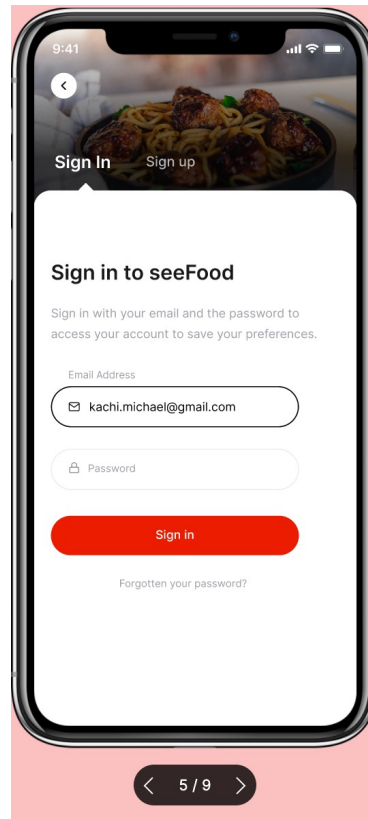
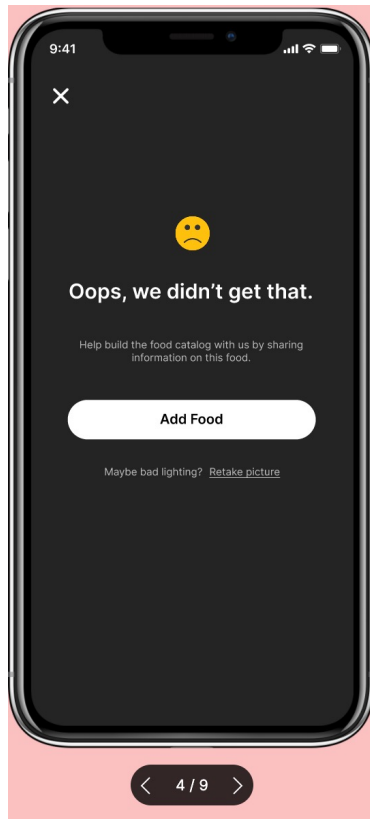
Design a draft interface of the app in software



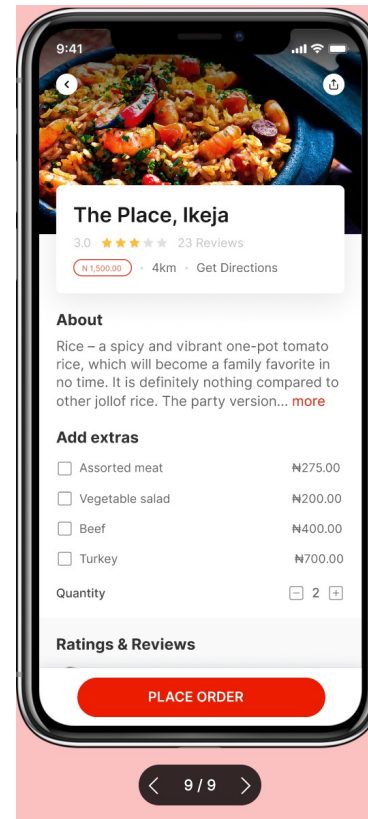
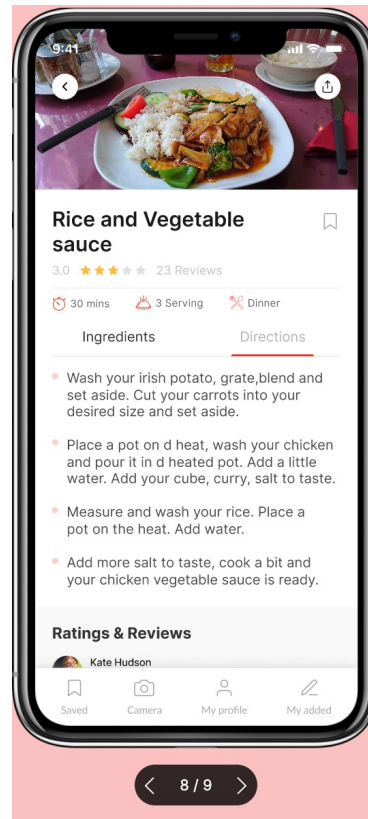
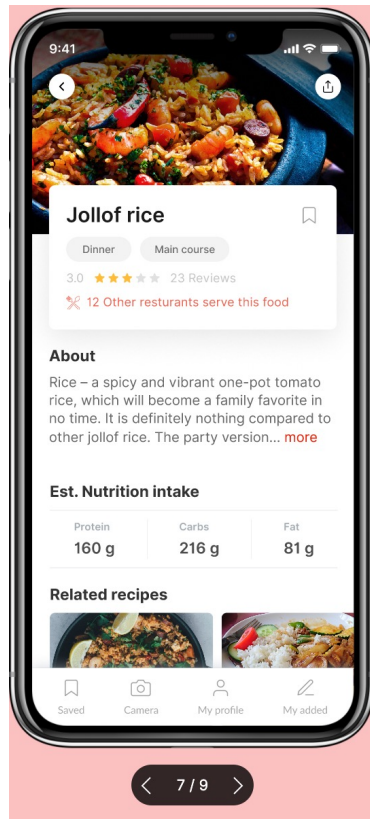
Final UI design to show the flow of the application



Final UI design to show the flow of the application (cont'd)



Final UI design to show the flow of the application (cont'd)



Summary

Storyboards allow **software development team** to easily and inexpensively

1. Describe user's problems
2. Visualise UX ideas
3. Visualise UI design

User Experience (UX) designers design wireframe prototype of the system

Graphic User Interface (UI) designers design all the visual elements

- buttons
- graphics
- color schemes
- typography
- etc.

Summary

Understand different types of page layout structure

- Linear
- Hierarchical
- Non-linear

Common types of UIs

- command line interface
- graphic user interface
- voice-enabled user interface

Reference

1. <https://softwarehut.com/blog/it-outsourcing/software-development-team-roles-and-responsibilities>
2. <https://www.uxbooth.com/articles/how-to-use-storyboards-for-product-development/>
3. <https://www.nngroup.com/articles/storyboards-visualize-ideas/>
4. <https://boords.com/storyboard-examples#animation-storyboards>
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7. <https://milanote.com/templates/storyboards/app-storyboard>
8. <https://ryanstutorials.net/software-design-and-development/storyboards.php>
9. https://youtu.be/5KUNmgt_pvY
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11. <https://medium.com/figma-africa/shazam-for-food-or-mobile-app-ui-design-process-2e358e853881>